Portfolio Risk, Sensitivity, and Constrained Allocation

Mathematical foundations, C++ implementation, and independent validation

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Abstract

This project develops an auditable numerical engine for portfolio risk, attribution, sensitivity, and contribution-only allocation. We derive the mathematical models, implement them in C++ with a scoped reverse-mode differentiation tape, and retain independent Python reference calculations. The central distinction is between arithmetic correctness, optimization correctness, and the empirical adequacy of estimated covariance. The paper states the domains of the derivatives, formulates the constrained quadratic problem, documents data assumptions, and separates measured validation from unperformed empirical work. Historical covariance is not treated as a known law of future returns. This is a research and educational software project.

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1. Definitions, prerequisites, and risk estimation

Practical question

Symbols used in this subsection

- i, t: Asset index and observation time; $t-1$ is the preceding observation.
- $r_{t,i}, r_t$: Simple return for asset i at time t, and vector of returns at that time.
- $P^*_{t,i}$: Consistently adjusted historical price; the star marks an adjustment label.
- $n, \mathbb{R}^n$: Number of assets and the space of real vectors with n entries.
- h_i, V, w_i, w: Dollar holding, total value, individual portfolio weight, and vector of weights.
- $\mathbf{1}, T$: All-ones vector, and transpose when used as a superscript.
- X, Y: Random variables in the expectation, variance, and covariance definitions.
- $E, \mathrm{Var}, \mathrm{Cov}$: Expectation, variance, and covariance operators.
- $\sum, \geq, >$: Summation, greater-than-or-equal comparison, and strict greater-than comparison.

How much dispersion does a fixed portfolio have under an estimated covariance model, and which assumptions enter that estimate?

For asset i, a simple one-period return is $r_{t,i} = P^*_{t,i}/P^*_{t-1,i} - 1$. Here P^* denotes a positive, consistently adjusted historical price series whose economic interpretation has been checked. This notation alone does not establish that dividends are reinvested. Current holdings require a separate unadjusted valuation price.

Let $r_t \in \mathbb{R}^n$, $w \in \mathbb{R}^n$, $w_i \geq 0$ and $\mathbf{1}^T w = 1$. Returns and weights are dimensionless. If h_i are dollar holdings, $V = \sum_i h_i > 0$ and $w_i = h_i/V$. Covariance is in squared-return units for a declared period; volatility is in return units for the same period.

Expectation is a probability-weighted average. Variance $\mathrm{Var}(X) = E[(X - E[X])^2]$ measures squared deviations around it. Covariance $\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$ measures joint deviations. Correlation divides covariance by the product of nonzero standard deviations. A zero-variance variable has no defined correlation coefficient under this formula.

Portfolio quadratic form

Symbols used in this subsection

- r, r_p: Asset-return vector and resulting portfolio return.
- w, w_i, w_j: Weight vector and its entries at asset indices i, j.
- $E, \mathrm{Var}, \sum_i, \sum_j$: Expectation, variance, and sums across asset indices.
- a: Arbitrary real vector used to test nonnegative quadratic variance.
- T: Transpose of a vector or matrix; not the observation count in this subsection.
- Q, Λ: Orthogonal eigenvector matrix and diagonal eigenvalue matrix.
- $\lambda_{max}, \lambda_{min}$: Largest and smallest eigenvalues; subscripts indicate extrema.

With fixed weights, $r_p = w^T r$. Expanding the square gives

$$\mathrm{Var}(r_p) = E\left[\left(\sum_i w_i(r_i - E[r_i])\right)^2\right] = \sum_i \sum_j w_i w_j \Sigma_{ij} = w^T \Sigma w.$$

Thus $\sigma(w) = \sqrt{w^T \Sigma w}$. Covariance is symmetric and positive semidefinite (PSD): $a^T \Sigma a = \mathrm{Var}(a^T r) \geq 0$ for every real a. The spectral theorem writes $\Sigma = Q \Lambda Q^T$; PSD means nonnegative eigenvalues. Eigenvalues measure variance along orthogonal directions, not individual asset risks.

Singularity means some direction has zero variance under this model. For positive-definite Σ, the spectral condition number is $\lambda_{max}/\lambda_{min}$; a large ratio warns that inversion can amplify perturbations. Risk evaluation needs matrix multiplication, not inversion. Singularity is therefore valid and must not be rejected simply because a Cholesky factorization fails. See MOSEK's portfolio formulation.

Sample estimator and normalization

Symbols used in this subsection

- T, n, t: Number of return observations, number of assets, and observation index.
- $r_t, \bar{r}$: Return vector at observation t and its sample mean.
- X: Centered return matrix, with observations as rows and assets as columns.
- S: Sample covariance matrix using the $T - 1$ denominator.
- a: Arbitrary test vector for the quadratic form.
- $^T, \|\cdot\|, \mathrm{rank}, \min$: Transpose, Euclidean norm, matrix rank, and minimum operator.
- $\mathbb{R}^{T \times n}, \sum_t$: Real matrices with T rows and n columns, and sum over observations.

For $T \geq 2$ aligned return rows, $\bar{r} = T^{-1} \sum_t r_t$ and centered matrix $X \in \mathbb{R}^{T \times n}$:

$$S = \frac{1}{T-1} X^T X.$$

This uses the sample normalization; under independent identically distributed rows with finite covariance it is unbiased. Rows of market data need not meet those assumptions. Since $a^T S a =$

$\|Xa\|^2/(T-1)$, this construction preserves PSD. Centering implies $\operatorname{rank}(S) \le \min(n, T-1)$. Pairwise deletion with different date sets need not preserve PSD; the product must use a common set of complete daily returns.

Example with three return rows $(.01, .02), (-.01, 0), (0, -.02)$ has zero mean and

$$S = \begin{pmatrix} .0001 & .0001 \\ .0001 & .0004 \end{pmatrix}.$$

The reference independently matches this result and NumPy's `cov(..., ddof=1)`.

Shrinkage and its precise status

Symbols used in this subsection

- S, S_{ML}, S_α: Sample covariance, empirical covariance using denominator T, and shrunk covariance.
- $\alpha, \hat{\alpha}$: Chosen shrinkage coefficient and fitted shrinkage coefficient; a hat marks estimation.
- μ, n, I: Mean diagonal variance, asset count, and identity matrix.
- $\operatorname{tr}, \lambda_i$: Sum of diagonal entries and eigenvalue indexed by i.
- $X, T, {}^T$: Centered return matrix, observation count, and transpose.

For a PSD estimate S, define $\mu = \operatorname{tr}(S)/n$ and

$$S_\alpha = (1-\alpha)S + \alpha\mu I, \qquad 0 \le \alpha \le 1.$$

It is PSD as a convex combination of PSD matrices. If $\alpha > 0$ and $\mu > 0$, all eigenvalues $(1-\alpha)\lambda_i + \alpha\mu$ are positive. This improves conditioning but introduces a structured bias toward equal variances and zero covariance; it is not a guarantee of better future allocation.

The engine implements a fixed-α sensitivity path. Separately, Python fits the Ledoit–Wolf scaled-identity estimator through scikit-learn and records the estimated coefficient. Its covariance-loss criterion is not realized portfolio returns. See the original paper and estimator documentation.

The fitted estimator uses the centered empirical covariance $S_{ML} = X^T X/T$, while the sample option uses $S = X^T X/(T-1)$. Tests reconstruct the fitted result as $(1-\hat{\alpha})S_{ML} + \hat{\alpha}\operatorname{tr}(S_{ML})I/n$. Both outputs are multiplied by the declared annualization factor. Thus estimator comparisons include a normalization difference as well as shrinkage. A fixed arbitrary coefficient is never labeled a fitted Ledoit–Wolf estimate.

Annualization and interpretation

Symbols used in this subsection

- K, d: Number of additive periods and a subscript identifying daily quantities.

For an additive K-period return sum, covariance equals $K\Sigma_d$ only if cross-period covariance terms vanish and period covariances are identical. With serial dependence the omitted cross terms matter. Annual volatility $\sqrt{252}\sigma_d$ is a convention under these assumptions, not the exact variance of

compounded annual simple returns. Display both period and multiplier. The example in chapter 2 is already annual and must not be annualized again.

This is a constant-weight risk estimate. A buy-and-hold historical account has drifting weights; reproducing that path requires holdings evolution. Historical covariance estimates future risk only through a modeling assumption of relevant history. The interface compares windows of 63, 126 and 252 completed common daily returns when available. It labels their spread sensitivity rather than calibrated uncertainty.

Implementation: reference.py and estimation.py. Validation includes hand values, scaling, permutation, PSD rejection, singular cases and estimator normalization. Exchange-calendar processing is implemented and tested with synthetic dates; empirical estimation quality is explored in chapter 5, without a claim of general predictive validity.

2. Differentiation and a worked example

From calculus to decisions

Symbols used in this subsection

- $w, v(w)$: Weight vector and variance evaluated at those weights.
- δ: Small vector perturbation to the weights.
- g, ∇, ∇_w: Volatility gradient, gradient operator, and differentiation with respect to weights.
- $d, \mathbf{1}$: Feasible transfer direction and all-ones vector.
- e_A, e_B: Unit vectors selecting assets A and B.
- ϵ, g_A, g_B: Small transfer size and gradient entries for assets A and B.
- $^T, O(\epsilon^2)$: Transpose and an error whose magnitude is bounded by a constant times the squared transfer size near zero.

For a fixed symmetric Σ, expand $(w+\delta)^T\Sigma(w+\delta)$ to obtain

$$v(w+\delta) = v(w) + 2\delta^T\Sigma w + \delta^T\Sigma\delta.$$

Therefore $\nabla v = 2\Sigma w$. Applying the scalar chain rule to $\sqrt{v}$ yields

$$g = \nabla_w \sigma = \frac{\Sigma w}{\sigma}, \qquad \sigma > 0.$$

The derivative is with respect to unconstrained coordinates even though valid portfolios lie on a simplex. A feasible first-order direction d must satisfy $\mathbf{1}^T d = 0$ and remain inside nonnegativity bounds. A transfer from A to B has $d = e_B - e_A$, so

$$\sigma(w+\epsilon d) - \sigma(w) = \epsilon(g_B - g_A) + O(\epsilon^2).$$

At a boundary a feasible direction may allow only one sign of ϵ. Do not confuse this derivative with increasing B while holding all other weights fixed.

Risk attribution

Symbols used in this subsection

- w, w_i, g: Weight vector, asset-i weight, and volatility-gradient vector.
- $RC_i, i, \sum_i$: Asset-i volatility contribution, asset index, and sum across assets.
- $^T, (\cdot)_i$: Transpose and extraction of vector entry i.

Volatility is homogeneous of degree one for positive scalings. Euler's identity gives $w^T g = \sigma$; directly,

$$\sum_i RC_i = \sum_i w_i \frac{(\Sigma w)_i}{\sigma} = \frac{w^T \Sigma w}{\sigma} = \sigma.$$

RC_i has the same units as volatility. Capital share w_i and risk share RC_i/σ answer different questions. Contributions can be negative, even in a long-only portfolio, if covariance gives a sufficiently strong hedge; they are an attribution, not independent stand-alone risks. The attribution framework is described in MOSEK's risk budgeting chapter.

New cash changes the denominator

Symbols used in this subsection

- h, h_i, H, V: Dollar holdings vector, holding of asset i, and two labels for existing total wealth.
- w, w_i, g, g_j: Weight vector, weight i, volatility-gradient vector, and gradient entry j.
- i, j, δ_{ij}: Asset indices and Kronecker delta: one for equal indices, zero otherwise.
- $\partial, \nabla_h, ^T, \sum_i$: Partial derivative, gradient with respect to holdings, transpose, and sum over assets.
- b, B, x: Transfer amount, new contribution budget, and purchase-dollar vector.
- g_A, g_B: Gradient entries for the two assets used in the transfer example.

For dollar holdings h, $H = \sum_i h_i > 0$, and $w = h/H$:

$$\frac{\partial w_i}{\partial h_j} = \frac{\delta_{ij} - w_i}{H}, \qquad \frac{\partial \sigma}{\partial h_j} = \frac{g_j - w^T g}{H} = \frac{g_j - \sigma}{H}.$$

The units are volatility per dollar. This derivative includes the effect on all weights. Adding dollars proportionally to every existing holding leaves weights unchanged: $h^T \nabla_h \sigma = 0$. A transfer of b dollars keeps total wealth constant and instead has derivative $b(g_B - g_A)/H$.

For a fixed contribution B fully distributed through purchases x, the feasible total is constant: $w(x) = (h + x)/(V + B)$. Its objective gradient in purchase coordinates is $2\Sigma(h + x)/(V + B)^2$. Off the budget constraint, that fixed-denominator extension is a computational definition, not the normalization for arbitrary additional wealth. Both give the same directional derivative along feasible transfers of purchases.

Hand-worked example

Symbols used in this subsection

- w, v, g, RC, h: Weights, portfolio variance, volatility gradient, risk-contribution vector, and dollar holdings.
- A, B: Labels for the first and second assets in this example; B here is not a contribution budget.

Take $w = (.6, .4)$ and annual $\Sigma = \begin{pmatrix} .04 & .006 \\ .006 & .01 \end{pmatrix}$. The asset volatilities are .2 and .1; the implied correlation is $.006/(.2 \cdot .1) = .3$.

$$\Sigma w = (.0264, .0076), \quad v = .6^2(.04) + 2(.6)(.4)(.006) + .4^2(.01) = .01888.$$

Thus $\sigma = 0.1374045123$, $g = (.1921334282, .0553111384)$, and $RC = (.1152800569, .0221244554)$. A accounts for 83.8983% of total volatility attribution despite holding 60% of capital.

A transfer of one percentage point of portfolio weight from A to B has linear change $-.0013682229$ and exact change $-.0013611369$ in volatility units. Multiply by 100 to express volatility percentage points: approximately $-.13682$ versus $-.13611$.

Start with $h = (600, 400)$ dollars and buy 100 of B using new cash. Weights become $(600/1100, 500/1100)$. Exact volatility change is $-.0072425026$ versus the local dollar approximation $-.0082093374$. A 10% wealth increase is large enough that the approximation error is visible. The conclusion concerns percentage-return volatility; dollar risk can increase when invested wealth grows.

Sensitivity to a valid covariance path

Symbols used in this subsection

- S, F, S_α: Base covariance, target covariance, and covariance along the shrinkage path.
- μ, I, α: Mean diagonal variance, identity matrix, and shrinkage coefficient.
- $w, ^T, \partial/\partial\alpha$: Weight vector, transpose, and partial differentiation with respect to shrinkage.

Keep base S, target $F = \mu I$, and weights fixed. On $S_\alpha = (1 - \alpha)S + \alpha F$:

$$\frac{\partial \sigma}{\partial \alpha} = \frac{w^T(F - S)w}{2\sigma}.$$

Only $\alpha \in [0, 1]$ is admitted. At endpoints use admissible one-sided perturbations for checks. This derivative does not differentiate the estimated Ledoit–Wolf coefficient with respect to data. Changing a single correlation arbitrarily can destroy PSD; use model-valid paths instead.

Implemented custom reverse mode

Symbols used in this subsection

- u, v, z: Two scalar inputs and their product in the tape example; v here is not portfolio variance.

- $\bar{u}, \bar{v}, \bar{z}$: Adjoints: derivatives of the final output with respect to these intermediate values.
- $+=$: Accumulate into an existing adjoint rather than overwrite it.
- $x, q, f, df/dx$: Scalar input, squared intermediate, final output, and derivative of the output with respect to the input.
- $m, n, O(\cdot)$: Operation count, asset count, and asymptotic upper-bound notation for computation or storage.

The C++ tape stores immutable node values and parent edges with local derivatives. Each reverse pass allocates fresh zero adjoints, initializes the output adjoint to one, and traverses backward to add chain-rule contributions. For $z = uv$, accumulate $\bar{u} \mathrel{+}= \bar{z}v$ and $\bar{v} \mathrel{+}= \bar{z}u$. Addition, subtraction, multiplication, division and square root are implemented. If $u = v$, both edge contributions accumulate. For example, $q = x^2$ and $f = q + q$ produce $df/dx = 4x$, including the shared intermediate correctly.

A tape owns node storage. Handles retain shared ownership and carry a generation number to reject cross-tape mixing and use after reset. Graphs are rebuilt per calculation, including every optimizer gradient evaluation; stored local derivatives are never reused with changed inputs. Division by zero, square root at nonpositive values and nonfinite arithmetic fail explicitly. The risk wrapper returns zero volatility with its usual gradient unavailable, even when a special constant-zero model admits a derivative. With m operations and bounded parents per node, forward evaluation and one reverse pass require $O(m)$ work and storage. The dense quadratic graph has $O(n^2)$ operations; no speed advantage over the closed form is asserted.

AD evaluates derivatives of the implemented arithmetic graph up to floating-point effects; it does not cure statistical model error. The closed-form volatility gradient is simple and remains an independent oracle. AD supplies implementation depth and handles composed normalization and parameter paths. See Baydin et al., section 3.

Independent formulas remain in reference.py. The C++ engine evaluates risk, normalized-dollar sensitivities and fixed-shrinkage sensitivities through the tape. Python tests compare these against analytical formulas and finite differences; native tests also exercise handle lifetime, reset and shared nodes under address and undefined-behavior sanitizers.

3. Constrained allocation

Practical question and domain

Symbols used in this subsection

- h, V, B, D: Existing dollar holdings, existing wealth, contribution budget, and final total wealth.
- x, x_i, c_i, u_i: Purchase vector, purchase for asset i, final-weight cap, and dollar purchase capacity.
- $n, i, \mathbf{1}, \mathbb{R}^n$: Asset count, asset index, all-ones vector, and real-vector space.
- $f, \min_x$: Portfolio variance objective and minimization over purchases.
- $^T, \nabla, \nabla^2, \succeq 0$: Transpose, gradient, Hessian (second-derivative matrix), and positive-semidefinite ordering.

Given nonnegative dollar holdings $h \in \mathbb{R}^n$, wealth $V = \mathbf{1}^T h$, contribution $B \geq 0$, and $D = V + B > 0$, distribute purchases x without selling. Assume a fixed PSD covariance, continuous dollar amounts, full investment of B and optional post-contribution caps $c_i \in [0, 1]$.

$$\min_x f(x) = \frac{(h+x)^T \Sigma (h+x)}{D^2}, \qquad \mathbf{1}^T x = B, \quad 0 \leq x_i \leq u_i = c_i D - h_i.$$

The objective has squared-return units; x and u are dollars. The gradient has squared-return units per dollar:

$$\nabla f(x) = \frac{2\Sigma(h+x)}{D^2}, \qquad \nabla^2 f = \frac{2\Sigma}{D^2} \succeq 0.$$

PSD curvature proves convexity: along every line the second derivative is nonnegative. The box intersected with the budget hyperplane is convex and compact when feasible. A continuous objective has a minimizer there. Strict positivity of curvature on nonzero feasible directions ensures uniqueness; singular covariance can instead produce many optimal allocations. Weight agreement is consequently an inadequate solver check.

Feasibility before optimization

Symbols used in this subsection

- $h, h_i, V, B, c, c_i, u, u_i$: Holdings vector and entry, existing wealth, budget, cap vector and entry, and purchase-capacity vector and entry.
- $i, \sum_i, \leq, \geq$: Asset index, sum across assets, less-than-or-equal, and greater-than-or-equal comparisons.

With upper bounds only, feasibility is equivalent to $u_i \geq 0$ for all i and $\sum_i u_i \geq B$. Necessity follows by summing upper bounds; sufficiency follows by filling nonnegative capacities until B is used. An existing overweight asset may become permissible after dilution by the contribution. Evaluate $h_i \leq c_i(V + B)$, not $h_i/V \leq c_i$.

For $h = (600, 400)$, $B = 200$, and $c = (.6, .6)$, $u = (120, 320)$ and the budget is feasible. Tightening the first cap to .4 makes its capacity -120, so no purchase-only solution exists. The reference implements these checks in `purchase_bounds`; the implemented solver's residual tolerances are specified below.

Algorithm and projection

Symbols used in this subsection

- x, x^k, x^{k+1}, x_i: Purchase vector, current iterate, next iterate, and purchase for asset i; k counts updates.
- C, Π_C: Feasible set and Euclidean projection onto that set.
- $u, B, h, c, D, \mathbf{1}$: Purchase capacities, budget, existing holdings, caps, final wealth, and all-ones vector.
- η, L, λ_{max}: Step size, gradient Lipschitz bound, and largest eigenvalue.

- $f, \nabla f, z, \tau$: Variance objective, its gradient, trial point, and scalar projection threshold.
- $\min, \max, \sum_i, \|\cdot\|, ^T$: Minimum, maximum, sum across assets, Euclidean norm, and transpose.
- ℓ, y, b, a, C_y: Existing holdings divided by final wealth, purchase fractions, budget fraction, capacity fractions, and their feasible set.
- s, Q, F: Largest covariance eigenvalue, covariance divided by that eigenvalue, and scaled variance objective.
- p, z_p, n: Pivot asset index, pivot score, and asset count in the extreme-range projection argument.
- $d, y^0, y^*, \nabla F$: Iterate displacement, initial purchase fractions, optimal purchase fractions, and scaled-objective gradient; star marks an optimum.

Projected gradient is scoped, transparent, and independently testable. Start from a feasible point and iterate

$$x^{k+1} = \Pi_C(x^k - \eta \nabla f(x^k)), \quad C = \{x : 0 \le x \le u,\ \mathbf{1}^T x = B\}.$$

For $L = 2\lambda_{max}(\Sigma)/D^2 > 0$, use $0 < \eta \le 1/L$ as an initial fixed-step design. If $L = 0$, every feasible point is optimal and no reciprocal step is formed. Projected gradient is the proximal-gradient method with an indicator of the feasible set; see Parikh and Boyd, section 4.2.

To project a trial point z, minimize $\frac{1}{2}\|x - z\|^2$ over C. Stationarity with an equality multiplier τ and box clipping yields

$$x_i(\tau) = \min\{u_i, \max\{0, z_i - \tau\}\}, \quad \sum_i x_i(\tau) = B.$$

The sum is continuous and nonincreasing. For $0 < B < \sum_i u_i$, bracket τ between $\min_i(z_i - u_i)$ and $\max_i z_i$, then bisect until the budget residual meets a documented scale-aware tolerance. The projected point is unique, though the multiplier need not be. Handle $B = 0$ and $B = \sum_i u_i$ directly. Validate projection feasibility, idempotence, and the variational inequality $(z - \Pi_C z)^T (y - \Pi_C z) \le 0$ for feasible y.

The implemented coordinates are $\ell = h/D$, $y = x/D$, $b = B/D$ and $a = c - \ell$. Thus $w = \ell + y$, with feasible set $C_y = \{y : 0 \le y \le a,\ \mathbf{1}^T y = b\}$. This avoids dollar-scale conditioning and remains defined when $B = 0$. For $s = \lambda_{max}(\Sigma) > 0$, define $Q = \Sigma/s$ and $F(y) = (\ell + y)^T Q(\ell + y)$. The gradient $2Q(\ell + y)$ has Lipschitz constant 2, so the code takes step $\eta = 1/2$. Each projection uses rescaled bisection for 160 iterations and a final budget-residual correction inside the box. The baseline projects equal purchase fractions onto the feasible set. Zero contribution or zero covariance has an immediate feasible optimum.

An extreme-range audit exposed a weakness in scaling by the full score range: a difference of order 10^{300} can erase a unit budget. The implementation now first replaces each upper bound by $\min(u_i, B)$, which leaves the feasible set unchanged. Sort scores in descending order and choose pivot p where their cumulative capped capacity first reaches B. At threshold z_p, purchased capacity is below B; at threshold $z_p - nB$, every asset at or above the pivot is saturated, giving capacity at least B. Hence a valid threshold lies in $[z_p - nB, z_p]$. After shifting by z_p and scaling by B, clip scores to $[-n, n]$: clipped extremes must be zero or saturated throughout this threshold interval.

Bisection therefore retains the budget geometry independently of the original score range. This argument concerns exact arithmetic; finite-precision residual correction and explicit diagnostics remain necessary.

To see why this step descends, let $d = y^{k+1} - y^k$. Projection optimality gives $\nabla F(y^k)^T d \le -\|d\|^2/\eta$. The Lipschitz-gradient bound then gives $F(y^{k+1}) \le F(y^k) - (1/\eta - L/2)\|d\|^2$. For $L = 2$ and $\eta = 1/2$, the coefficient is one. In exact arithmetic, convex projected gradient also satisfies $F(y^k) - F(y^*) \le \|y^0 - y^*\|^2/(2\eta k)$ for $k \ge 1$. This is an objective bound, not a guarantee of unique weights. Floating-point projection error and a finite iteration limit make explicit diagnostics necessary.

Optimality and independent validation

Symbols used in this subsection

- $f, x, u, \mathbf{1}$: Dollar-coordinate variance objective, purchase vector, capacities, and all-ones vector.
- ν, λ, γ: Budget-equality multiplier, nonnegative lower-bound multipliers, and nonnegative upper-bound multipliers.
- λ_i, γ_i, i: Individual constraint multipliers and asset index; λ here denotes multipliers, not covariance eigenvalues.
- $\eta, G_\eta, \Pi_C, \nabla f$: Step size, projected-gradient mapping, feasible-set projection, and objective gradient.
- $y, C_y, F, s, \|\cdot\|_2$: Purchase fractions, their feasible set, scaled variance objective, covariance scale, and Euclidean norm.

For the original dollar problem, KKT stationarity is

$$\nabla f + \nu\mathbf{1} - \lambda + \gamma = 0,$$

with unrestricted ν, nonnegative λ, γ, feasibility, $\lambda_i x_i = 0$ and $\gamma_i(x_i - u_i) = 0$. For this convex quadratic on a polyhedron, an appropriate first-order variational inequality characterizes global optima. A convenient numerical diagnostic is

$$G_\eta(x) = \frac{x - \Pi_C(x - \eta\nabla f(x))}{\eta}.$$

It vanishes exactly at a constrained optimum. The implementation stops when the scaled mapping norm $2\|y - \Pi_{C_y}(y - \nabla F(y)/2)\|_2$ is at most 10^{-9} by default, or after 10,000 updates. This absolute, dimensionless threshold is adjustable. Multiplying this norm by s gives the mapping norm in weight coordinates for the unscaled variance objective with step $1/(2s)$; it is not a dollar-coordinate norm. The output also reports dollar budget residual, maximum weight-bound violation, iteration count, stopping reason, objective and improvement over the feasible baseline. Constraint feasibility admits only 10^{-12} weight-scale roundoff before projection. Independent tests compare full-rank and rank-deficient objectives with OSQP, emphasizing objective agreement instead of unique weight agreement.

The C++ solver obtains the composed objective gradient through the tape by default; an analytical-gradient mode is also checked. Covariance stays fixed within each solve. The interface repeats solves

across predeclared windows and covariance estimators when enough observations exist. Synthetic convergence and solver-agreement evidence is reported in chapter 5, alongside a selected historical-market allocation stability study. That study is descriptive and does not establish future stability.

Minimum estimated variance ignores expected return, investor objectives, tax, fees, execution and estimation error. It answers this explicit constrained mathematical question; it does not define the best personal investment portfolio.

4. Architecture and data methodology

Implemented separation of responsibilities

The dependency chain is provider adapter, validated returns, Python covariance estimation, C++ risk/AD/optimization, then Streamlit presentation. The C++17 engine accepts arrays and explicit parameters without network or dashboard dependencies. CMake and scikit-build-core package a pybind11 extension. Eigen supplies dense arithmetic and symmetric eigendecomposition. The independent NumPy reference remains separate.

Bindings validate numerical inputs in the core and translate exceptions. Returned arrays own their Python-visible storage. The allocator releases the Python interpreter lock while computing. Result metadata includes engine version, estimation method, normalization, window, annualization, dataset hash and diagnostics. These fields describe reproducibility inputs, not immutable historical data.

Provider adapter and limits

The Alpaca adapter loads credentials server-side from environment variables or Streamlit secrets. Only market-data endpoints are called. Historical requests explicitly select IEX or SIP and adjustment mode `all`; pagination follows page tokens, rejecting repeated tokens and excessive pagination. Authentication and malformed responses fail clearly. Transport errors, HTTP 429 and server failures receive at most four attempts with bounded waits. There is no silent feed fallback. Alpaca's access documentation distinguishes coverage and subscription requirements.

The adapter has a bounded, 15-minute in-memory history cache, isolated per dashboard session and credential identity. Cache keys include symbols, range, feed and adjustment mode. Requests end at least 15 minutes before analysis time rather than in the future. On expiry or manual refresh the entire adjusted window is fetched, and snapshots are compared for added, removed and revised observations, including older dates. Returned histories are defensive copies. This audits incremental changes without mixing adjustment vintages; it does not reduce network volume. An overlap-only refresh cannot guarantee detection of arbitrary revisions outside its overlap.

Historical bars documentation lists adjustment choices, but an option name alone does not establish dividend reinvestment economics. Known split/dividend cases, account entitlement and display/redistribution permissions remain unverified. Checked-in fixtures are synthetic. Mocked HTTP tests establish software behavior, not successful live access. For an independently specified corporate event, the audit helper compares the adjusted-price return with the holder's wealth return

$$r_{\text{event}} = \frac{kP_{\text{after}} + d}{P_{\text{before}}} - 1,$$

where k is new shares per old share and d is cash per old share.

- r_{event}: Holder wealth return across the event; event is a label.
- k, d: New shares per old share, and cash distribution per old share.
- $P_{\text{before}}, P_{\text{after}}$: Share prices before and after the event.

A two-for-one split from 100 to 50 gives zero wealth return. A cash dividend of one with prices 100 and 99 also gives zero. Backward price adjustment is not generally identical to cash-wealth arithmetic when the ex-date price also changes for other reasons. Tests intentionally detect that discrepancy; the helper cannot certify a provider convention without checked live events.

Calendar and observation policy

Only XNYS sessions whose close plus a 15-minute publication allowance precedes the analysis time enter daily-return preparation. Explicit UTC timestamps are converted to New York session dates. Prices must be positive and finite; duplicate observations fail. Missing history at the latest completed session fails. The selected universe is USD US stocks and unleveraged ETFs; user confirmation is required, while automated asset eligibility validation remains pending.

Prices are reindexed to the expected calendar before returns are computed. Each return requires prices on adjacent expected sessions, so a missing Monday cannot turn Friday-to-Tuesday movement into a daily return. Only then are common complete return rows retained. Effective row counts, excluded dates and a SHA-256 identity of the prepared returns are recorded. No forward-filling occurs. Tests cover gaps, weekends, incomplete sessions, pagination, retries, staleness and estimator normalization.

Current shares use separately retrieved, unadjusted latest trades. A price older than 15 minutes during an open session, or older than 15 minutes before the last completed close while closed, is flagged stale. Prices more than 30 seconds into the future and nonpositive prices are rejected. Stale share valuation stops the interface. A last trade is an observation, not an executable quote; this threshold is an explicit policy, not a provider guarantee. Dollar and hypothetical-weight inputs do not need the valuation request.

Interface and deployment boundary

The Streamlit app supports a credential-free synthetic example, attribution, exact versus local purchase effects, contribution allocation and diagnostics. Market mode adds dollar/share/weight input, feed selection and comparisons over sample and Ledoit–Wolf estimates at 63, 126 and 252 observations when available. Zero-volatility gradients and infeasible caps are explained explicitly.

On 9 October 2026, the private Streamlit deployment successfully imported and ran the C++ engine. Its synthetic workflow reproduced 13.74% volatility and the 200-dollar purchase of B. Authenticated IEX retrieval returned 379 common completed daily observations for SPY/BND without dropped return dates. The hypothetical holdings of 600/400 dollars gave 8.4781% estimated annual volatility using 252 observations; allocation of a 200-dollar contribution gave 7.4276%. All six window/estimator comparisons converged. The Share dialog showed the public-access option unchecked. These are hosted integration checks at source revision `15ccd273`, separate from Ubuntu tests. Credentials stayed in server secrets; no brokerage execution was performed.

5. Numerical verification and reproducibility

Evidence and reproduction

Symbols used in this subsection

- w, w^*, v: Feasible allocation, optimal allocation, and linear-oracle minimizing allocation; v here is a vector, not variance.
- $f, \nabla f, ^T$: Convex allocation objective, its gradient, and transpose.

At revision `fb1d449b`, run 38088543928 passed all 44 tests on Python 3.10 and 3.12, native sanitizers and all report commands. The added audit checks 48 randomized risk instances, 48 allocation solves against OSQP and an independent linear-oracle convex certificate, covariance/dollar scales from 10^{-200} to 10^{200}, extreme score differences, negative risk contributions and nullspace portfolios. It also checks exact historical-prefix invariance and independently known cash-flow ledgers. For a feasible w and a linear-oracle minimizer v of $\nabla f(w)^T v$, convexity yields

$$0 \le f(w) - f(w^*) \le \nabla f(w)^T (w - v).$$

The oracle fills the available capacity in increasing gradient order; it does not reuse the projection routine. These bounded tests support the implemented behavior, not a proof of reliability for every floating-point input.

Install a C++17 compiler, CMake and Eigen3, then run from the repository root:

```
python -m pip install '.[test,app]'
python -m unittest discover -s tests -v
python -m examples.two_asset --output work/two-asset.json
python -m examples.engine_report --output work/engine-report.json
```

The C++ report script records synthetic data identity, source hashes, runtime environment, derivative errors and allocation diagnostics. The Python 3.12.15 / NumPy 2.5.3 run measured maximum weight-gradient error $2.78 \cdot 10^{-17}$ and dollar-gradient error $4.07 \cdot 10^{-20}$ against the independent reference. The worked allocation had zero reported budget, bound and projected-gradient residuals. These values describe this fixture, not worst-case error bounds. Each Python job also stores JSON artifacts. Source hashes identify the evaluated code; later documentation edits do not change the historical record.

Test family	Independent evidence
Risk and attribution	Hand variance .01888; NumPy formula; Euler identity
Gradients	Analytical, AD and central-difference comparisons
Dollar normalization	Recomputed holdings and feasible transfer derivatives
Tape semantics	Shared nodes, repeated passes, lifetime, reset and domains
Projection	Budget and bounds, idempotence, variational inequality

Test family	Independent evidence
Allocation	OSQP objectives for full-rank and singular matrices
Degenerate cases	Zero budget/covariance, iteration limit, infeasible caps
Covariance	PSD validation, rank deficiency, scaling and permutation
Estimation	Hand covariance and reconstructed fitted shrinkage
Data	Mock pagination/retries, calendar gaps, staleness and caching
Interface	Synthetic display, infeasible caps, universe confirmation

Native tests additionally run with AddressSanitizer and UndefinedBehaviorSanitizer. Passing this finite set provides evidence for covered cases, not a proof of memory safety for every input. Tests use synthetic inputs without provider credentials.

Tolerances and interpretation

Symbols used in this subsection

- $n, \epsilon_{mach}, S_{ij}$: Asset count, machine epsilon, and covariance entry at row i, column j.
- $h, O(h^2), O(\epsilon_{mach}/h)$: Finite-difference step (not holdings here), truncation-error order, and cancellation-error order.
- $\max, |\cdot|$: Maximum operator and absolute value.

Calculations use float64. Covariance validation tolerates asymmetry and negative eigenvalues only at $100n\epsilon_{mach} \max |S_{ij}|$. It symmetrizes and clips negative eigenvalues only inside that allowance, then uses the same effective matrix for risk and gradients. Materially indefinite matrices are rejected. This relative rule has no absolute floor hiding errors at small scales. The risk API reports the Frobenius norm of its covariance correction.

Weights must sum to one within 10^{-12}; they are not silently renormalized. Dollar normalization is explicit. Central finite differences have truncation error $O(h^2)$ and cancellation error roughly $O(\epsilon_{mach}/h)$. Multiple step sizes are used, with relative tolerances up to $3 \cdot 10^{-6}$ for coarse steps and absolute allowances near zero. See gradcheck mechanics; PyTorch is not a dependency.

OSQP checks request absolute and relative tolerances 10^{-10}. Comparisons allow objective relative error 10^{-7} and absolute error 10^{-10}, alongside engine convergence and feasibility diagnostics. Singular problems may have several minimizers, so equal weights are not required. Near-zero volatility can make sensitivities ill-conditioned even when variance evaluation is stable. Extreme floating-point scale testing remains a hardening task.

Worked interpretation

Symbols used in this subsection

- g_j, V, j: Volatility-gradient entry, total portfolio value, and asset index.

The synthetic portfolio has volatility .1374045123 and contributions .1152800569 and .0221244554. Fixed-alpha shrinkage changes volatility from .1374045 at zero to .1140175 at one; this spread measures model dependence, not statistical confidence. The purchase example shows increasing error in a local derivative approximation as purchases grow.

For the same covariance, holdings $(600, 400)$, contribution 200 and caps $(.6, .6)$, the solution purchases $(0, 200)$ and ends at $(.5, .5)$ with variance .0155. Along the feasible interval the variance derivative favors additional B, but the no-sale bound prevents reducing A below .5. The optimum is at a boundary despite a nonzero unconstrained gradient.

Historical evaluation

Historical contribution backtest

- h_{t-1}, h_t^-, h_t, x_t: Prior end-of-day holdings, holdings after deposits before returns, current end-of-day holdings, and current purchase dollars; minus labels a pre-return state.
- t, r_t, R_t: Day index, vector of asset returns, and portfolio return excluding the direct deposit effect.
- $\mathbf{1}, ^T, \odot$: All-ones vector, transpose, and entrywise multiplication.
- $I_t, I_T, I_0, \prod_t$: Time-weighted wealth index, its final value, initial index of one, and product over days.
- $T, s, I_s, \min_t, \max$: Final observation index, earlier day index in the running maximum, index at that earlier day, minimum over days, and maximum operator.
- $s(R_t)$: Sample standard deviation of the daily portfolio-return series; s is a function name here, not the earlier covariance scale.

The cumulative evaluator is separate from the fixed-holdings variance study. At each decision date it fits Ledoit–Wolf covariance on the preceding 126 observations, before consuming that day's return. Initial holdings are 1,000 dollars divided equally; every 21 observations adds 200 dollars. There are no sales and no concentration cap below 100%. The comparator makes equal-dollar purchases on the same dates with the same cash flows. Neither policy chooses parameters from future outcomes.

Let $h_t^- = h_{t-1} + x_t$ be holdings after any contribution but before the daily return. Then $h_t = h_t^- \odot (\mathbf{1} + r_t)$ and the portfolio return is $R_t = \mathbf{1}^T h_t / \mathbf{1}^T h_t^- - 1$. The time-weighted index $I_T = \prod_t (1 + R_t)$ removes the direct deposit effect. Drawdown is $\min_t \{I_t / \max_{0 \le s \le t} I_s - 1\}$ with $I_0 = 1$; annualized volatility is $\sqrt{252}\, s(R_t)$. Terminal wealth is reported separately because it includes deposits. Holdings drift daily; this differs from the constant-weight portfolio used in variance calibration.

Zero returns must give terminal wealth equal to initial wealth plus deposits, zero time-weighted return and zero drawdown. A common 1% daily return has a closed-form wealth ledger, independently tested. Changing observations after a cutoff must leave the earlier cash-flow ledger and decisions exactly unchanged. Contiguous input arrays fix layout-dependent reduction differences without weakening that exact-prefix test. Missing market return dates suppress the hosted backtest, since skipping them would omit unknown wealth changes. Fees, taxes, spreads, executable prices, point-in-time vintages and survivorship effects are not modeled.

The seeded four-regime Gaussian stress simulation contains 1,008 observations and four assets. It evaluates 882 observations and 42 contributions totaling 8,400 dollars. On the Python 3.10 runner, optimized / equal-purchase terminal wealth was 8,460.41 / 8,507.02 dollars, time-weighted return 1.3306% / 2.3450%, volatility 23.3873% / 23.2907%, and drawdown −38.0320% / −37.0318%. The optimized policy was worse on these realized measures: minimizing fitted variance at each contribution is not a guarantee of lower subsequent variance, drawdown or higher return. This is synthetic stress evidence, not market performance.

The extended hosted market-data run used SPY, BND, GLD and QQQ with 1,513 complete IEX-adjusted daily returns, retrieved on 9 October 2026 at 22:05:14 UTC under revision `cb737b9b`. The backtest evaluated 1,387 observations from 5 April 2021 through 9 October 2026, making 67 contributions totaling 13,400 dollars. All solves converged; no missing daily returns were skipped. Aggregate results were:

Measure	Minimum variance	Equal purchases
Terminal wealth (dollars)	17,738.59	23,078.07
Time-weighted return	31.99%	93.75%
Annualized volatility	7.20%	12.27%
Maximum drawdown	−18.66%	−21.70%

Lower volatility and drawdown accompanied lower return and wealth on this chosen history. This tradeoff does not establish general superiority. The independent fixed-holdings study on the same snapshot completed 360 model evaluations across 60 test blocks. Mean calibration ratios for sample / Ledoit–Wolf were 1.1684 / 1.1366 (63 observations), 1.1975 / 1.1765 (126), and 1.1870 / 1.1696 (252). Ratios above one indicate underestimation of subsequent variance on average in these groups. Present-day adjustments, surviving selected tickers and the unverified provider dividend convention prevent treating this as an unbiased investment backtest. These aggregates describe a recorded historical snapshot, not a fresh-data guarantee. The live report script reproduces the evaluation protocol against retrieved history; raw market histories and credentials are not committed.

Variance calibration and limitations

- w_t, r_s, t, s: Weights fitted at decision time t, asset returns in later test observation s, decision index, and test-observation index.
- $\mathrm{Var}_{\mathrm{test}}, {}^{T}$: Sample variance within the held-out test block and transpose.
- L^1: Sum of absolute component changes in consecutive allocation vectors; the superscript identifies the norm, not a power.

Unperformed work includes live corporate-action validation, SIP entitlement checks and broader empirical analysis. The implemented research protocol fits windows of 63, 126 and 252 observations through a cutoff, then evaluates the next 21 observations, using nonoverlapping test blocks. Each cutoff poses an independent contribution problem with fixed initial holdings of 100 dollars per asset and a 200-dollar contribution. It is not a cumulative trading simulation. For fitted weights w_t, it compares predicted annual variance with $252\,\mathrm{Var}_{\mathrm{test}}(r_s^T w_t)$, using sample normalization within the test block. The calibration ratio and successive allocation L^1 changes summarize variance fit and stability. Future-perturbation tests ensure that changing test observations cannot change the first fitted allocation. Present-day adjusted history and a surviving-symbol universe can still introduce vintage or selection bias despite correct slicing.

The hosted runner at revision `8da7a371` completed 36 evaluations across six test blocks on the SPY/BND snapshot retrieved at `2026-10-09T21:38:30.377874+00:00`. Mean calibration ratios were:

Training observations	Sample	Ledoit–Wolf
63	0.9040	0.8344
126	1.0251	0.9666
252	1.2418	1.1551

Each group contains only six evaluations. Mean successive allocation L^1 change was zero in each group, consistent with unchanged boundary solutions. This selected two-asset example cannot establish estimator superiority or general allocation stability. The daily-return dataset SHA-256 identity is

```
2a3c00006e0e1b8b35309c375b0c00989
d291c180d78ca569382c371e9905080
```

The benchmark compares validated risk APIs against the vectorized NumPy reference, with five warm-up calls and seven batches of 30 calls at 2, 10 and 50 assets. It records median and percentile timings, compiler, platform, NumPy build configuration and thread settings. These are end-to-end costs including validation, eigendecomposition and Python/C++ crossing, not isolated kernel timings. Isolating pure binding overhead remains a separate benchmark. API timings do not imply any investment-return advantage.

Measured timings

The synthetic factor study uses seed 24109, 504 observations and three assets. It completed 72 evaluations with mean calibration ratio 1.0096. This is a reproducible exercise of the protocol under synthetic data, not empirical evidence about market covariance or an estimated confidence level. Full evaluation records are retained in the workflow artifacts.

On the Python 3.12 Ubuntu x86-64 runner, using GCC 13.3 and one configured BLAS/OpenMP thread, measured median microseconds per validated API call were:

Assets	NumPy	C++ analytical	C++ AD
2	34.44	3.48	4.48
10	45.14	9.98	30.41
50	285.10	182.34	884.69

The analytical C++ path was faster for these cases, while the AD path was slower than NumPy at 50 assets. Graph construction and traversal have a material cost. This result depends on the tested APIs, runner and dimensions; it is not a claim that C++ or AD is universally faster. The benchmark script records percentile timings and the numerical-library environment.

Comprehension questions

Symbols used in this subsection

- g_j, V, j: Volatility-gradient entry, total portfolio value, and asset index.

1. Why does a 60% capital holding contribute about 84% of volatility?

2. Why is a new-dollar derivative $(g_j - \sigma)/V$ instead of g_j/V?
3. Why can singular covariance be valid while optimal weights are nonunique?
4. Why can a constrained optimum have a nonzero ordinary gradient?
5. Which assumptions justify square-root-of-time annualization?

6. References and glossary

Sources consulted on 2026-10-08. Documentation is mutable; recheck when implementing integrations. Entries distinguish material actually consulted from full proofs not yet reviewed.

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3. **scikit-learn**, LedoitWolf API. Consulted the scaled-identity formula and centering contract. The estimator wrapper is implemented and empirical normalization is tested.
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6. **PyTorch**, Gradcheck mechanics. Consulted central finite differences and analytical comparison. Reference tests do not use PyTorch.
7. **pybind11**, official documentation. Consulted C++/Python integration overview. The extension uses CMake and scikit-build-core; tests exercise arrays and handle lifetime.
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Glossary

Term	Meaning here
Covariance	Expected product of centered returns
PSD	Every quadratic form is nonnegative
Eigenvalue	Variance scaling along an eigenvector direction
Conditioning	Sensitivity of a numerical problem to input perturbations
Shrinkage	Blend an estimate with a structured target
Gradient	Vector of unconstrained coordinate derivatives
Directional derivative	Local change along a specified direction
Adjoint	Derivative of the output with respect to an intermediate node
Tape	Stored computational graph and intermediate values
Projection	Nearest feasible point in the chosen norm
KKT conditions	Feasibility, stationarity, multiplier signs and complementarity
Residual	Quantified violation of a constraint or optimality condition
Annualization	Conversion under declared time-aggregation assumptions
Sensitivity	Change under specified perturbations, not a confidence interval